

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	G. Santosh Kumar	Reg. No.:	192524020
Section / Batch:	2025 Batch	Date:	24/07/2026

Code of Conduct

I G. Santosh Kumar _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Santosh Kumar _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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My Assessments

Course: CSA0305RA

PSEUDOCODE WRITING TASK
OPEN

CO5_AT1_Pseudocode Writing Task
0/5 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
OPEN
✓ Completed

DEBUGGING & OPTIMIZATION TASK
10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
OPEN
✓ Completed

DEBUGGING & OPTIMIZATION TASK
10/10 answered

Review →

CO3_AT3_MCQ
OPEN

Multiple Choice Questions - Non Linear Data Structures
0/25 answered

Attempt →

CO1_AT1_C CODE OUTPUT PREDICTION
OPEN
✓ Completed

Review →

1925241

Kaggle

CSA030

CSA030

Deep D

Relation

Relation

Downlo

SIMATS

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5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int key) {
4     int low = 0, high = n-1;
5     while(low<=high){
6         int mid = low + (high-low) /2;
7
8         if(arr[mid] == key)
9             return mid;
10    }
```

Test Input

stdin input (optional)

Output

Element Found 3

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CSA030

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3 int stack[MAX] = {10,20,30};
4 int top = -1;
5 int pop() {
6     if(top == -1){
7         printf("Underflow");
8         return -1;
9     }
10    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Singly Linked List
5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int stack[MAX];
5 int top;
6 void init() {
7     top = -1;
8 }
9
10 int main() {
11     init();
12     printf("top = %d",top);
13 }
```

Test Input

stdin input (optional)

Output

top = -1

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Singly Linked List
5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int stack[MAX];
5 int top;
6 void init() {
7     top = -1;
8 }
9
10 int main() {
11     init();
12     printf("top = %d",top);
13 }
```

Test Input

stdin input (optional)

Output

top = -1

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) ← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q5 Stack - Balancing Symbols 5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[]')) {  
    pop();  
} else {  
    return 0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[100];  
4 int top = -1;  
5  
6 void pop() {  
7     if(top != -1) {  
8         top--;  
9     }  
10 }
```

Test Input

(

Output

Enter Brackets:
Brackets does not match

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) ← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q6 Singly Linked List 5 marks

What is the issue?

```
temp = head  
while temp.next != NULL:  
    print(temp.data)  
    temp = temp.next
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node{  
5     int data;  
6     struct Node *next;  
7 };  
8  
9 void display(struct Node *head) {  
10     struct Node *temp;  
11     temp = head;  
12     while(temp != NULL) {  
13         printf("%d", temp->data);  
14     }
```

Test Input

stdin input (optional)

Output

102030

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q7 Stack5 marks

Point out the error.
POP(stack);
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int stack[MAX] = {10,20,30};
5 int top = 2;
6 int pop() {
7     if(top == -1){
8         printf("Underflow");
9         return -1;
10    } return stack[top--];
11 }
```

Test Input

stdin input (optional)

Output

Deleted element = 30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8 Linked List5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node{
5     int data;
6     struct Node *next;
7 };
8
9 void reverse(struct Node *head) {
10     struct Node *prev = NULL, *current = head,*next;
11
12     while(current != NULL){
13         next = current -> next;
```

Test Input

stdin input (optional)

Output

Head after reverse = 30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search
5 marks

Q2 Stack
5 marks

Q3 Stack
5 marks

Q4 Stack
5 marks

Q5 Stack - Balancing Symbols
5 marks

Q6 Singly Linked List
5 marks

Q7 Stack
5 marks

Q9 Linked List

5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node{
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *new;
12
13     new = (struct Node *)malloc(sizeof(struct Node));
14     new->data = 10;
15 }
```

Test Input

stdin input (optional)

Output

10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack
5 marks

Q2 Circular Queue
5 marks

Q3 Stack
5 marks

Q4 Stack
5 marks

Q5 Doubly Linked List
5 marks

Q6 Queue
5 marks

Q7 Queue - Deque
5 marks

Q1 Stack

5 marks

Point out the error?
PEEK(stack);
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 100
5 struct Stack {
6     int arr[MAX];
7     int top;
8 };
9
10 void initStack(struct Stack* s) {
11     s->top = -1;
12 }
13
14 int isFull(struct Stack* s) {
15     return s->top == MAX - 1;
16 }
```

Test Input

stdin input (optional)

Output

Pushed 10 onto the stack.
Pushed 20 onto the stack.
Pushed 30 onto the stack.
The top element (peek) is: 30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)Back

QUESTIONS

Q1Stack5 marks

Q2Circular Queue5 marks

Q3Stack5 marks

Q4Stack5 marks

Q5Doubly Linked List5 marks

Q6Queue5 marks

Q7Queue - Deque5 marks

Q2Circular Queue5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1;
7 int rear = -1;
8
9 void enqueue(int vehicle) {
10     if ((rear + 1) % MAX == front) {
11         printf("Queue Full\n");
12     } else {
13         rear = (rear + 1) % MAX;
14         queue[rear] = vehicle;
15     }
16 }
```

Test Input

stdin input (optional)

Output

```
Vehicle 101 added to the signal.
Vehicle 102 added to the signal.
Vehicle 103 added to the signal.
Vehicle 104 added to the signal.
Vehicle 105 added to the signal.
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)Back

QUESTIONS

Q1Stack5 marks

Q2Circular Queue5 marks

Q3Stack5 marks

Q4Stack5 marks

Q5Doubly Linked List5 marks

Q6Queue5 marks

Q7Queue - Deque5 marks

Q3Stack5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 5
5
6 int stack[MAX];
7 int top = -1;
8
9 void push(int value) {
10     if (top == MAX - 1) {
11         printf("Overflow\n");
12     } else {
13         top++;
14         stack[top] = value;
15     }
16 }
```

Test Input

stdin input (optional)

Output

```
Overflow
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)← Back

QUESTIONS

Q1Stack5 marks

Q2Circular Queue5 marks

Q3Stack5 marks

Q4Stack5 marks

Q5Doubly Linked List5 marks

Q6Queue5 marks

Q7Queue - Deque5 marks

Q4 Stack5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

✳️ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 100
5
6 int stack[MAX];
7 int top = -1;
8
9 void push(int value) {
10     if (top == MAX - 1) {
11         return;
12     }
13     top++;
14     stack[top] = value;
15 }
```

Test Input

stdin input (optional)

Output

```
20
10
Underflow
-1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)← Back

QUESTIONS

Q1Stack5 marks

Q2Circular Queue5 marks

Q3Stack5 marks

Q4Stack5 marks

Q5Doubly Linked List5 marks

Q6Queue5 marks

Q7Queue - Deque5 marks

Q5 Doubly Linked List5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
 if (del->prev != NULL)
 del->prev->next = del->next;
 if (del->next != NULL)
 del->next->prev = del->prev;
}

✳️ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* prev;
7     struct Node* next;
8 };
9
10 void deleteNode(struct Node** head_ref, struct Node* del) {
```

Test Input

stdin input (optional)

Output

```
Created DLL: 40 30 20 10
DLL after deleting node 30: 40 20
10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) ← Back

QUESTIONS

Q1 Stack 5 marks ✓

Q2 Circular Queue 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Doubly Linked List 5 marks ✓

Q6 Queue 5 marks ✓

Q7 Queue - Deque 5 marks ✓

Q6 Queue 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 100
5
6 int queue[MAX];
7 int front = -1;
8 int rear = -1;
9
10 void enqueue(int data) {
11     if (rear == MAX - 1) {
12         return;
13     }
```

Test Input

stdin input (optional)

Output

```
10 20 30
10
20 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) ← Back

QUESTIONS

Q1 Stack 5 marks ✓

Q2 Circular Queue 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Doubly Linked List 5 marks ✓

Q6 Queue 5 marks ✓

Q7 Queue - Deque 5 marks ✓

Q7 Queue - Deque 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 5
5
6 int deque[MAX];
7 int front = -1;
8 int rear = -1;
9
10 void insertFront(int data) {
11     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
12         printf("Deque Overflow\n");
13     }
```

Test Input

stdin input (optional)

Output

```
30 20 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1

Stack

5 marks

Q2

Circular Queue

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l <= r) {
5         int mid = l + (r - l) / 2;
6
7         if (arr[mid] == x)
8             return mid;
9
10        if (arr[mid] > x)
```

Test Input

stdin input (optional)

Output

Element is present at index 3

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1

Stack

5 marks

Q2

Circular Queue

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 5
5
6 int queue[MAX];
7 int front = -1;
8 int rear = -1;
9
10 void enqueue(int value) {
```

Test Input

stdin input (optional)

Output

10 20 30 40 50
Dequeued: 10
Dequeued: 20
30 40 50

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue - Deque 5 marks

Q10 Stack - Postfix Expression 5 marks

Find the bug in the evaluation of a Postfix expression.
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <ctype.h>
4
5 #define MAX 100
6
7 int stack[MAX];
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q10 Queue 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int Q[MAX];
5 int front = -1;
6 int rear = -1;
7
8 int DEQUEUE() {
9 if (front == -1 || front > rear) {
10 printf("Underflow");
11 return -1;
12 }
13 else {
```

Test Input

5
10 20 30 40 50

Output

Underflow-1